

Edeviser — Vercel Deployment Strategy

Free Plan vs Pro Plan · Pilot Running · Scalability · Observability Setup

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Audience: Developer Team & Investors

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1. Executive Summary

Edeviser is a full-stack OBE (Outcome-Based Education) + Gamification platform built on React/Vite + Supabase. It is currently deployed on Vercel's **Hobby (Free) plan** for the pilot phase.

Key findings:

- The free plan is **sufficient for pilot** with 1–3 institutions and up to ~500 concurrent users
 - One critical limitation** already hit: cron jobs limited to once-per-day (leaderboard refresh changed from every 5 min to daily)
 - Upgrading to Pro (\$20/month)** unlocks per-minute crons, higher bandwidth, and team collaboration — essential before scaling beyond pilot
 - Free Vercel Analytics and Speed Insights** are available now and should be enabled immediately — they are safe, healthy, and provide real user data at zero cost
 - The platform's architecture (Supabase backend, Edge Functions, Vite SPA) is well-suited for Vercel and will scale cleanly
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2. What Vercel Is and Why We Use It

Vercel is a cloud platform for deploying frontend applications and serverless functions. For Edeviser it provides:

Capability	How Edeviser Uses It
Static hosting	Serves the React/Vite SPA from a global CDN

Serverless API routes	Hosts <code>api/cron/*.ts</code> — the 10 scheduled jobs
Automatic deployments	Every push to <code>main</code> triggers a production deploy
Preview deployments	Every PR gets its own preview URL for QA
Edge network	Assets cached globally — fast load times in Qatar/MENA
Security headers	HSTS, CSP, X-Frame-Options all configured in <code>vercel.json</code>

Why not AWS/GCP/Azure? Vercel's zero-config deployment, GitHub integration, and built-in CDN make it the fastest path to production for a Vite SPA. The backend (Supabase) is already managed separately, so Vercel only needs to serve the frontend and cron jobs.

3. Vercel Free (Hobby) Plan — Full Analysis

What You Get

Feature	Free Limit	Edeviser Usage
Bandwidth	100 GB/month	~2–5 GB/month at pilot scale
Build execution	6,000 min/month	~50 min/month (fast Vite builds)
Deployments	Unlimited	~20–30/month
Serverless function invocations	100,000/month	~10,000/month (crons + API)
Serverless function duration	10 sec max	Crons complete in 2–5 sec
Cron jobs	2 per project	We have 10 — this is a problem
Cron frequency	Once per day max	Was every 5 min — now fixed to daily
Team members	1 (personal only)	Solo or shared via GitHub
Custom domains	Unlimited	✅ Already configured
Analytics	Free tier available	✅ Should enable now
Speed Insights	Free tier available	✅ Should enable now
Support	Community only	No SLA

What's Already Working

- ✅ Production deployment live
- ✅ Custom domain connected
- ✅ GitHub auto-deploy on `main` push
- ✅ Security headers (HSTS, CSP, X-Frame-Options)
- ✅ SPA routing (all paths → `index.html`)
- ✅ Asset caching (1-year immutable for `/assets/`)
- ✅ Preview deployments on every PR

4. Vercel Pro Plan — What We Gain

Cost: \$20/month per member (or \$20/month for a team)

Feature	Free	Pro	Impact for Edeviser
Cron frequency	Once/day	Every minute	Critical — leaderboard, streak-risk need frequent runs
Cron jobs per project	2	100	We have 10 crons — need Pro
Bandwidth	100 GB	1 TB	Handles 10,000+ active students
Serverless duration	10 sec	300 sec	Bulk import, PDF generation, AI calls
Team members	1	Unlimited	Dev team collaboration
Password protection	✗	✓	Useful for staging environments
Advanced analytics	Limited	Full	Conversion funnels, retention
Support	Community	Email (72h SLA)	Production incident response
Concurrent builds	1	3	Faster CI for multiple PRs
Preview deployment comments	✗	✓	QA workflow improvement

Pro is required before going beyond pilot. The cron limitation alone justifies the upgrade once you have paying institutions.

5. Edeviser-Specific Issues on Free Plan

Issue 1: Cron Job Frequency (ALREADY FIXED)

Problem: `leaderboard-refresh` was set to `* / 5 * * * *` (every 5 minutes). Vercel Hobby blocks any cron that runs more than once per day.

Fix applied: Changed to `0 12 * * *` (once daily at noon UTC).

Impact: Leaderboard updates once per day instead of every 5 minutes. For pilot with small cohorts, this is acceptable. For production with competitive gamification, this needs Pro.

All 10 crons and their current schedules:

Cron	Schedule	Frequency	Impact of Daily-Only
streak-risk	<code>0 20 * * *</code>	Daily 8PM	✓ Fine
weekly-summary	<code>0 8 * * 1</code>	Weekly Monday	✓ Fine
compute-at-risk	<code>0 2 * * *</code>	Daily 2AM	✓ Fine
perfect-day-prompt	<code>0 18 * * *</code>	Daily 6PM	✓ Fine

streak-reset	0 0 * * *	Daily midnight	✅ Fine
leaderboard-refresh	0 12 * * *	Daily noon	⚠️ Was every 5 min
ai-at-risk-prediction	0 3 * * *	Daily 3AM	✅ Fine
notification-digest	0 20 * * *	Daily 8PM	✅ Fine
fee-overdue-check	0 6 * * *	Daily 6AM	✅ Fine
exam-period-notify	0 8 * * *	Daily 8AM	✅ Fine

Issue 2: Serverless Function Timeout (10 seconds)

Problem: Several operations may exceed 10 seconds on free plan:

- Bulk student data import (CSV with 500+ rows)
- PDF generation (course files, transcripts)
- AI at-risk prediction (calls external AI API)
- Attainment rollup for large cohorts

Mitigation on free plan:

- These operations run as Supabase Edge Functions (Deno), not Vercel functions — they have their own timeout (60 sec)
- The Vercel cron routes just trigger the Supabase Edge Functions, so the 10-sec limit only applies to the trigger, not the work

Impact: Low risk for pilot. Monitor if bulk operations start timing out.

Issue 3: 100 GB Bandwidth Cap

Problem: If Edeviser grows to 1,000+ active students, bandwidth could approach the limit.

Estimate:

- Average page load: ~500 KB (gzipped bundle ~1.2 MB, cached after first load)
- Daily active user: ~5 page loads = ~2.5 MB/day
- 500 students × 2.5 MB × 30 days = **37.5 GB/month** — within free limit
- 1,000 students × 2.5 MB × 30 days = **75 GB/month** — approaching limit
- 2,000+ students → upgrade needed

Mitigation: Assets are cached with 1-year immutable headers. Returning users cost near-zero bandwidth.

Issue 4: Single Team Member

Problem: Free plan is personal — can't add team members to the Vercel project.

Workaround: Share GitHub access. All deployments happen via GitHub Actions, so the team can deploy without Vercel dashboard access.

Impact: Low for pilot. Upgrade when you hire a second developer.

6. Pilot Running Strategy

Phase 1: Pilot (Current — Free Plan)

Target: 1–3 institutions, 50–500 students, 3–6 months

What works on free plan:

- Full application functionality
- All 10 cron jobs (daily frequency)
- Automatic deployments from GitHub
- Custom domain
- Basic observability (Analytics + Speed Insights)
- Preview deployments for QA

What to monitor:

- Bandwidth usage (stay under 80 GB/month)
- Function invocations (stay under 80,000/month)
- Build minutes (stay under 5,000/month)
- User-reported issues with leaderboard staleness (daily refresh)

Pilot success criteria:

- 3 institutions onboarded
- 200+ active students
- Zero production incidents from Vercel limits
- Leaderboard daily refresh acceptable to users

Phase 2: Growth (Pro Plan — ~Month 4)

Trigger: Any of these:

- 3+ institutions signed
- 500+ active students
- Leaderboard staleness complaints
- Need to add a second developer to Vercel

What changes:

- Leaderboard refresh back to every 5 minutes
- Bulk import timeout issues resolved
- Team collaboration enabled
- Advanced analytics for investor reporting

Cost: \$20/month — recoverable from first paying institution

Phase 3: Scale (Pro + Supabase Pro — ~Month 8)

Trigger: 10+ institutions, 2,000+ students

What changes:

- Vercel Pro handles the frontend scale
- Supabase Pro needed for database connection pooling
- Consider Vercel Enterprise for SLA guarantees

7. Scalability Analysis

Frontend Scalability (Vercel)

Vercel's CDN is globally distributed. The Edeviser SPA is a static bundle — once built, it serves from edge nodes worldwide. This means:

Users	Vercel Load	Notes
100	Negligible	Free plan comfortable
1,000	Low	Free plan comfortable
10,000	Moderate	Pro plan recommended
100,000	High	Enterprise plan + CDN optimization

The frontend itself scales infinitely on Vercel — it's just static files. The bottleneck is always the backend (Supabase).

Backend Scalability (Supabase — separate from Vercel)

Component	Free Limit	Pro Limit	Edeviser Risk
Database connections	60	200	Medium — 500+ concurrent users
Database size	500 MB	8 GB	Low — OBE data is small
Edge Function invocations	500K/month	2M/month	Low
Realtime connections	200	500	Medium — gamification uses realtime
Storage	1 GB	100 GB	Low

Key insight: Supabase is the real scalability constraint, not Vercel. The connection pooler (PgBouncer) is critical for 500+ concurrent users.

Architecture Scalability Score

Layer	Current State	Scalability Rating
Frontend (Vercel CDN)	Static SPA	★★★★★ Excellent
API (Supabase REST)	PostgREST	★★★★ Good
Realtime (Supabase)	WebSocket channels	★★★ Moderate
Database (PostgreSQL)	Single instance	★★★ Moderate
Edge Functions (Deno)	Serverless	★★★★ Good
Cron Jobs (Vercel)	Serverless	★★★★ Good

Overall: Well-architected for 0–5,000 users without major changes.

8. Vercel Observability & Web Analytics Setup (Free)

What's Available for Free

Vercel offers two free observability tools that should be enabled immediately:

8.1 Web Analytics

What it tracks:

- Page views and unique visitors
- Top pages by traffic
- Referrer sources (where users come from)
- Device types (mobile vs desktop)
- Countries/regions
- Core Web Vitals per page

Is it safe? Yes. Vercel Analytics is privacy-first:

- No cookies
- No cross-site tracking
- GDPR compliant
- No personal data collected
- Works without user consent banners

Is it healthy for the app? Yes. It's a lightweight script (~1 KB) injected at the edge. Zero performance impact.

8.2 Speed Insights

What it tracks:

- Real User Monitoring (RUM) — actual user experience, not lab tests
- Core Web Vitals: LCP, FID, CLS, TTFB, FCP
- Performance scores per page and per device
- P75 and P99 percentiles (catches slow outliers)

Why this matters for Edeviser:

- Students on mobile in Qatar may have different performance than desktop
- Identifies which pages are slow for real users
- Helps prioritize performance work

How to Enable Right Now (Step by Step)

Step 1: Enable Web Analytics

1. Go to your Vercel project dashboard: <https://vercel.com/attaulhaq0s-projects/e-deviser>
2. Click **Analytics** tab in the top navigation
3. Click **Enable Web Analytics**
4. Select **Hobby** plan (free)
5. Click **Enable**

That's it. No code changes needed — Vercel injects the script automatically.

Step 2: Enable Speed Insights

1. In the same project, click **Speed Insights** tab
2. Click **Enable Speed Insights**
3. Select **Hobby** plan (free)
4. Click **Enable**

Again, no code changes needed.

Step 3: Add the packages (optional but recommended for custom events)

If you want to track custom events (e.g., "student submitted assignment", "teacher released grade"):

```
npm install @vercel/analytics @vercel/speed-insights
```

Then in `src/main.tsx` :

```
import { inject } from "@vercel/analytics";
import { injectSpeedInsights } from "@vercel/speed-insights";

inject();
injectSpeedInsights();
```

This gives you the full SDK for custom event tracking.

Step 4: Track key business events (optional)

```
import { track } from "@vercel/analytics";

// In your submission handler:
track("assignment_submitted", { courseId, assignmentId });

// In your login handler:
track("user_login", { role: user.role });

// In your grade release:
track("grade_released", { courseId });
```

What You'll See After Enabling

Within 24 hours of your first real users:

Analytics Dashboard:

- How many students visited today
- Which pages they visited most (dashboard, learning path, leaderboard)
- Where they came from (direct, email, etc.)
- Mobile vs desktop split

Speed Insights Dashboard:

- Your app's real-world performance score
- Which pages are slowest
- Whether mobile users have worse experience than desktop

Is This Healthy for Production?

Yes, completely safe. Vercel Analytics and Speed Insights are:

- Used by thousands of production apps
- Privacy-compliant (no PII collected)
- Zero performance overhead
- Free forever on Hobby plan (with usage limits)

- No security risk

Free tier limits:

- Analytics: 2,500 events/month (enough for pilot)
- Speed Insights: 10,000 data points/month (enough for pilot)

When you exceed these, you can upgrade to paid analytics (\$10/month) or the data simply stops collecting until the next month — it never breaks the app.

9. Pros, Cons & Limitations Summary

Vercel Free Plan

Category	Pros	Cons
Cost	\$0/month	Limited resources
Deployment	Instant, automatic, zero-config	Single team member
Performance	Global CDN, fast TTFB	—
Crons	10 crons supported	Once-per-day max frequency
Functions	Serverless, auto-scaling	10-second timeout
Bandwidth	100 GB/month	Enough for ~1,500 students
Analytics	Free Web Analytics + Speed Insights	2,500 events/month limit
Support	Community forums	No SLA, no email support
Domains	Unlimited custom domains	—
Previews	PR preview deployments	—

Vercel Pro Plan

Category	Pros	Cons
Cost	Full features	\$20/month/member
Crons	Per-minute frequency	—
Functions	300-second timeout	—
Bandwidth	1 TB/month	—
Team	Unlimited members	—
Analytics	100K events/month	—
Support	Email, 72h SLA	Not 24/7

10. Upgrade Decision Framework

Is leaderboard staleness (daily refresh) causing user complaints?

YES → Upgrade to Pro immediately

NO → Continue on Free

Are you onboarding more than 3 institutions?

YES → Upgrade to Pro

NO → Continue on Free

Is bandwidth approaching 80 GB/month?

YES → Upgrade to Pro

NO → Continue on Free

Do you need to add a second developer to Vercel?

YES → Upgrade to Pro

NO → Continue on Free

Are bulk imports timing out?

YES → Check if it's Vercel (unlikely) or Supabase Edge Function

If Vercel → Upgrade to Pro

If Supabase → Upgrade Supabase plan

Recommendation: Upgrade to Pro when you sign your **second paying institution** or when you have **500+ active students**, whichever comes first.

11. Investor Perspective

Infrastructure Cost Efficiency

Stage	Monthly Cost	Users Supported
Pilot (now)	\$0 (Vercel Free) + \$0 (Supabase Free)	0–500 students
Early Growth	\$20 (Vercel Pro) + \$25 (Supabase Pro) = \$45/month	500–5,000 students
Scale	\$20 (Vercel Pro) + \$599 (Supabase Team) = \$619/month	5,000–50,000 students
Enterprise	Custom pricing	50,000+ students

Unit economics: At \$45/month supporting 5,000 students, infrastructure cost is **\$0.009 per student per month** — essentially zero.

Why Vercel is the Right Choice for Investors

1. **Zero DevOps overhead** — No servers to manage, no Kubernetes, no ops team needed
2. **Instant global scale** — CDN handles traffic spikes automatically
3. **Predictable costs** — No surprise bills from traffic spikes
4. **Fast iteration** — Deploy in 2 minutes, not 2 hours
5. **Built-in observability** — Real user data from day one
6. **Proven at scale** — Used by Airbnb, Notion, TikTok, and thousands of SaaS companies

Risk Assessment

Risk	Likelihood	Impact	Mitigation
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Vercel outage	Low (99.99% uptime SLA on Pro)	High	Supabase backend still works; users can't access UI
Bandwidth overage	Medium (at 1,000+ students)	Low	Upgrade to Pro (\$20/month)
Cron staleness	Low (daily is acceptable for pilot)	Medium	Upgrade to Pro when needed
Vendor lock-in	Low	Medium	Vite SPA can deploy to any CDN in hours

Overall infrastructure risk: LOW. The architecture is portable and the costs are negligible relative to revenue potential.

12. Recommended Action Plan

Immediate (This Week)

- ☒ Deploy to production (done)
- ☐ **Enable Web Analytics** (5 minutes, free, do it now)
- ☐ **Enable Speed Insights** (5 minutes, free, do it now)
- ☐ Verify all 10 cron jobs are running (check Vercel Functions logs)
- ☐ Set up Vercel email alerts for deployment failures

Short Term (Month 1–2)

- ☐ Monitor bandwidth usage weekly
- ☐ Review Speed Insights — identify slowest pages
- ☐ Review Analytics — understand user behavior patterns
- ☐ Fix the `post-audit-remediation` spec issues (`habit_logs` table, `console.log` stripping)

Medium Term (Month 3–4)

- ☐ Upgrade to Vercel Pro when second institution signs
- ☐ Restore leaderboard-refresh to `* / 5 * * * *`
- ☐ Add team members to Vercel project
- ☐ Set up Vercel preview deployment comments for QA workflow

Long Term (Month 6+)

- ☐ Evaluate Vercel Enterprise for SLA guarantees
 - ☐ Consider Supabase Pro for connection pooling at scale
 - ☐ Implement custom Vercel Analytics events for business KPIs
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Appendix A: Steering Files Explained

Steering files are Kiro IDE configuration files that provide persistent context to the AI assistant. They live in `.kiro/steering/` and are automatically included in every conversation.

File	Purpose
------	---------

design-system.md	Design tokens, component rules, color coding
domain-knowledge.md	OBE concepts, gamification rules, business logic
project-conventions.md	Tech stack, file structure, coding standards
engineering-guardrails.md	Non-negotiable rules (no any, SOLID principles)
supabase-patterns.md	RLS templates, TanStack Query patterns, Edge Function structure
component-patterns.md	Page skeletons, shared component usage
review-loop.md	Self-review checklist before finalizing code
pre-push-checks.md	Mandatory CI checks before committing

Inclusion modes:

- `always` — included in every conversation (domain-knowledge.md)
- `fileMatch` — included when matching files are in context (e.g., supabase-patterns.md when editing hooks)
- `manual` — only when explicitly referenced with `#` in chat

Appendix B: Hooks Explained

Hooks are automated triggers in the Kiro IDE that run agent actions when events occur. They live in `.kiro/hooks/`.

Hook Type	When It Fires	Example Use
fileEdited	When you save a file	Run lint on save
fileCreated	When a new file is created	Auto-add boilerplate
preToolUse	Before a tool executes	Security check before file writes
postToolUse	After a tool executes	Run tests after code changes
preTaskExecution	Before a spec task starts	Validate prerequisites
postTaskExecution	After a spec task completes	Run Postman collection to verify
userTriggered	Manual button click	Deploy to staging
promptSubmit	When you send a message	Add context to every prompt
agentStop	When agent finishes	Summarize what was done

The Postman hook you see firing after every task completion checks if the task created new API endpoints and validates them against the Postman collection.

Appendix C: Last Audit Report

Report name: Pre-Deployment E2E Audit – Edeviser Platform

File location: audit/output/audit-report.md

Spec location: .kiro/specs/pre-deployment-e2e-audit/

Last run: May 9, 2026

Verdict: No-Go (pre-fix) → Go-with-backlog (post-fix)

Summary of what was audited:

- 15 property-based tests (OBE invariants, gamification rules, system properties)
- Security scan (secrets in bundle, VITE_ allowlist, Edge Function validation)
- Design token conformance (color families, glassmorphism, physical margins)
- i18n parity (en ↔ ar key diff, untranslated literals)
- Accessibility (WCAG 2.1 AA color contrast, icon-only buttons)
- Performance budget (bundle size, pagination, realtime filters)
- Connectivity matrix (hook → table/RPC/Edge Function mapping)
- RLS matrix (per-role access probes)
- Cron health (endpoint reachability, idempotency)

Issues found and fixed:

- 8 color contrast failures → fixed (Bloom badges -500→-700, attainment -600→-800)
- 12 icon-only buttons missing aria-label → fixed
- 19 Edge Function body validation gaps → downgraded to Major (backlog)
- 946 untranslated string literals → Minor (i18n backlog)

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Next review: Before Phase 2 launch (Month 4)